

Vector Calculus

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(adapted from Fieller (2016), [Section 7.2](#))

This section covers derivatives of functions of vectors and matrices. Linear algebra prerequisites — including vectors, matrices, transpose, dot product, and quadratic forms — are covered in [Linear Algebra](#).

Let $\tilde{x}$ and $\tilde{\beta}$ be column vectors of length p (see [column vector](#) and [dot product](#)).

Definition 0.1 (Vector derivative). If $f(\tilde{\beta})$ is a scalar-valued function of a $p \times 1$ vector $\tilde{\beta}$, such as $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$, then its **vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) \\ \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) \\ \vdots \\ \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

Definition 0.2 (Row-vector derivative). If $f(\tilde{\beta})$ is a scalar-valued function of a $p \times 1$ vector $\tilde{\beta}$, such as $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$, then its **row-vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) & \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) & \cdots & \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

Theorem 0.1 (Row and column derivatives are transposes).

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) &= \left(\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) \right)^\top \\ \frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) &= \left(\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) \right)^\top \end{aligned}$$

i Proof

Proof. By Definition 0.1 and Definition 0.2, entry j of both $\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta})$ and $\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta})$ is $\frac{\partial}{\partial \beta_j} f(\tilde{\beta})$; the first is a $p \times 1$ column and the second a $1 \times p$ row with the same entries in the same order, so each is the transpose of the other. $\square$

Example 0.1 (Row and column derivatives of a linear function). For $f(\tilde{\beta}) = 3\beta_1 + 5\beta_2$:

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad \frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = [3 \quad 5],$$

and each is the transpose of the other.

Definition 0.3 (Derivative of a vector-valued function). If $\tilde{y} = \tilde{y}(\tilde{\beta}) = (y_1, \dots, y_q)^\top$ is a $q \times 1$ vector-valued function of the $p \times 1$ vector $\tilde{\beta}$, its **derivative with respect to $\tilde{\beta}$** is the $p \times q$ matrix whose (i, j) entry is

$$\left[\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top \right]_{ij} \stackrel{\text{def}}{=} \frac{\partial}{\partial \beta_i} y_j, \quad i = 1, \dots, p, \quad j = 1, \dots, q.$$

These notes use this *denominator layout* throughout: rows index the entries of $\tilde{\beta}$ (the denominator) and columns index the entries of $\tilde{y}$ (the numerator), so column j is the vector derivative $\frac{\partial}{\partial \tilde{\beta}} y_j$ (Definition 0.1). Both $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}$ and $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top$ denote this $p \times q$ matrix.

Some sources use the transposed, *numerator layout*, in which the derivative is the $q \times p$ Jacobian matrix with (j, i) entry $\frac{\partial}{\partial \beta_i} y_j$. Check a source's layout before combining its formulas with these.

Example 0.2 (Differentiating a 3×1 function of a 2×1 vector). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$ ($p = 2$) and $\tilde{y}(\tilde{\beta}) = (\beta_1^2, \beta_1\beta_2, 3\beta_2)^\top$ ($q = 3$). Then

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top}_{2 \times 3} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} \beta_1^2 & \frac{\partial}{\partial \beta_1} \beta_1\beta_2 & \frac{\partial}{\partial \beta_1} 3\beta_2 \\ \frac{\partial}{\partial \beta_2} \beta_1^2 & \frac{\partial}{\partial \beta_2} \beta_1\beta_2 & \frac{\partial}{\partial \beta_2} 3\beta_2 \end{bmatrix} = \begin{bmatrix} 2\beta_1 & \beta_2 & 0 \\ 0 & \beta_1 & 3 \end{bmatrix}$$

Definition 0.4 (Constant). A $q \times 1$ vector $\tilde{x}$ is **constant with respect to** the $p \times 1$ vector $\tilde{\beta}$ if its derivative (Definition 0.3) is zero:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{p \times q} = \mathbf{0}_{p \times q}$$

Example 0.3 (A constant vector). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$ and $\tilde{x} = (3, 5)^\top$, so $x_1 = 3$ and $x_2 = 5$ do not depend on $\tilde{\beta}$. Expanding $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top$ into its matrix of scalar partial derivatives (Definition 0.3) and evaluating each entry:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{2 \times 2} = \frac{\partial}{\partial \tilde{\beta}} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} x_1 & \frac{\partial}{\partial \beta_1} x_2 \\ \frac{\partial}{\partial \beta_2} x_1 & \frac{\partial}{\partial \beta_2} x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} 3 & \frac{\partial}{\partial \beta_1} 5 \\ \frac{\partial}{\partial \beta_2} 3 & \frac{\partial}{\partial \beta_2} 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{0}_{2 \times 2}$$

Every entry is the derivative of a constant, so $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top = \mathbf{0}_{2 \times 2}$ and $\tilde{x}$ is constant with respect to $\tilde{\beta}$ (Definition 0.4).

Theorem 0.2 (Derivative of a dot product). *If $\tilde{x}$ is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{x} \cdot \tilde{\beta})}_{p \times 1} = \underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta} \cdot \tilde{x})}_{p \times 1} = \tilde{x}_{p \times 1}$$

i Proof

Proof.

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}} (\tilde{x} \cdot \tilde{\beta}) &= \begin{bmatrix} \frac{\partial}{\partial \beta_1} (x_1 \beta_1 + x_2 \beta_2 + \dots + x_p \beta_p) \\ \frac{\partial}{\partial \beta_2} (x_1 \beta_1 + x_2 \beta_2 + \dots + x_p \beta_p) \\ \vdots \\ \frac{\partial}{\partial \beta_p} (x_1 \beta_1 + x_2 \beta_2 + \dots + x_p \beta_p) \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \\ &= \tilde{x} \end{aligned}$$

□

Example 0.4 (Derivative of a dot product). Let $\tilde{x} = (3, 5)^\top$ (constant with respect to $\tilde{\beta}$; see Example 0.3) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{x} \cdot \tilde{\beta} = 3\beta_1 + 5\beta_2$, and by Theorem 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta})}_{2 \times 1} = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Verifying entry-wise:

$$\frac{\partial}{\partial \tilde{\beta}}(3\beta_1 + 5\beta_2) = \begin{pmatrix} \frac{\partial}{\partial \beta_1}(3\beta_1 + 5\beta_2) \\ \frac{\partial}{\partial \beta_2}(3\beta_1 + 5\beta_2) \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Both methods agree.

Theorem 0.3 (Product rule for dot-products). *If $\tilde{a} = \tilde{a}(\tilde{x})$ and $\tilde{b} = \tilde{b}(\tilde{x})$ are differentiable $p \times 1$ vector functions of $\tilde{x}$, then:*

$$\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \cdot \underbrace{\tilde{b}}_{p \times 1} = \left(\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{a}^\top}_{1 \times p} \right) \underbrace{\tilde{b}}_{p \times 1} + \left(\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{b}^\top}_{1 \times p} \right) \underbrace{\tilde{a}}_{p \times 1}$$

i Proof

Proof. Entry-wise, for $i = 1, \dots, p$:

$$\begin{aligned} \left[\frac{\partial}{\partial \tilde{x}}(\tilde{a} \cdot \tilde{b}) \right]_i &= \frac{\partial}{\partial x_i} \sum_{k=1}^p a_k b_k \\ &= \sum_{k=1}^p \left(b_k \frac{\partial}{\partial x_i} a_k + a_k \frac{\partial}{\partial x_i} b_k \right) \\ &= \left[\left(\frac{\partial}{\partial \tilde{x}} \tilde{a}^\top \right) \tilde{b} \right]_i + \left[\left(\frac{\partial}{\partial \tilde{x}} \tilde{b}^\top \right) \tilde{a} \right]_i \end{aligned}$$

□

Example 0.5 (Example of the dot-product rule). Apply Theorem 0.3 with the vector $\tilde{\beta} = (\beta_1, \beta_2)^\top$ in the role of $\tilde{x}$. Let $\tilde{a}(\tilde{\beta}) = (\beta_1, \beta_1\beta_2)^\top$ and $\tilde{b}(\tilde{\beta}) = (\beta_2, \beta_1)^\top$. Then:

$$\tilde{a} \cdot \tilde{b} = \beta_1 \cdot \beta_2 + \beta_1\beta_2 \cdot \beta_1 = \beta_1\beta_2 + \beta_1^2\beta_2$$

By direct calculation:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{a} \cdot \tilde{b})}_{2 \times 1} = \frac{\partial}{\partial \tilde{\beta}}(\beta_1\beta_2 + \beta_1^2\beta_2) = \begin{pmatrix} \beta_2 + 2\beta_1\beta_2 \\ \beta_1 + \beta_1^2 \end{pmatrix}$$

By the product rule (Theorem 0.3), using $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{a}^\top}_{2 \times 2} = \begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}$ and $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{b}^\top}_{2 \times 2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$:

$$\begin{aligned} \underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{a} \cdot \tilde{b})}_{2 \times 1} &= \underbrace{\begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_2 \\ \beta_1 \end{pmatrix}}_{2 \times 1} + \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}}_{2 \times 1} \\ &= \begin{pmatrix} \beta_2 + \beta_1 \beta_2 \\ \beta_1^2 \end{pmatrix} + \begin{pmatrix} \beta_1 \beta_2 \\ \beta_1 \end{pmatrix} \\ &= \begin{pmatrix} \beta_2 + 2\beta_1 \beta_2 \\ \beta_1^2 + \beta_1 \end{pmatrix} \end{aligned}$$

Both methods agree.

Theorem 0.4 (Derivative of a linear map). *If $\mathbf{A}$ is an $m \times p$ matrix that is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{\beta})}_{p \times m} = \underbrace{\mathbf{A}^\top}_{p \times m}$$

i Proof

Proof. For entry (i, j) , where row i indexes the denominator $\tilde{\beta}$ (see Definition 0.3) and column j indexes the numerator $\mathbf{A} \tilde{\beta}$:

$$\begin{aligned} \left[\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{\beta}) \right]_{ij} &= \frac{\partial}{\partial \beta_i} (\mathbf{A} \tilde{\beta})_j \\ &= \frac{\partial}{\partial \beta_i} \sum_{k=1}^p a_{jk} \beta_k \\ &= a_{ji} \\ &= [\mathbf{A}^\top]_{ij} \end{aligned}$$

□

Example 0.6 (Derivative of a linear map). Let $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix} (1 \times 2)$ and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{A} \tilde{\beta} = 2\beta_1 + 3\beta_2$, and by Theorem 0.4:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{\beta})}_{2 \times 1} = \underbrace{\mathbf{A}^\top}_{2 \times 1} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Theorem 0.5 (Vector-derivative of a matrix-vector product). *If $\mathbf{A}$ is an $m \times q$ matrix that is constant with respect to $\tilde{\beta}$, and $\tilde{v} = \tilde{v}(\tilde{\beta})$ is a $q \times 1$ vector that depends on the $p \times 1$ vector $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{v})}_{p \times m} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{A}^\top}_{q \times m}$$

This result generalizes Theorem 0.4, which is the special case $\tilde{v} = \tilde{\beta}$ (so that $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}$ and $\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{\beta}) = \mathbf{A}^\top$).

i Proof

Proof. For entry (i, j) , where row i indexes the denominator $\tilde{\beta}$ and column j indexes the numerator $\mathbf{A}\tilde{v}$ (see Definition 0.3):

$$\begin{aligned} \left[\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{v}) \right]_{ij} &= \frac{\partial}{\partial \beta_i}(\mathbf{A}\tilde{v})_j \\ &= \frac{\partial}{\partial \beta_i} \sum_{k=1}^q a_{jk} v_k \\ &= \sum_{k=1}^q a_{jk} \frac{\partial}{\partial \beta_i} v_k \\ &= \sum_{k=1}^q \left[\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right]_{ik} [\mathbf{A}^\top]_{kj} \\ &= \left[\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{A}^\top \right]_{ij} \end{aligned}$$

□

Example 0.7 (Vector-derivative of a matrix-vector product). Let $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$ (1×2 , constant) and $\tilde{v}(\tilde{\beta}) = (\beta_1^2, \beta_2^2)^\top$. Then $\mathbf{A}\tilde{v} = 2\beta_1^2 + 3\beta_2^2$. By Theorem 0.5:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{v})}_{2 \times 1} = \begin{pmatrix} 2\beta_1 & 0 \\ 0 & 2\beta_2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4\beta_1 \\ 6\beta_2 \end{pmatrix}$$

Theorem 0.6 (Vector-derivative of a product of matrices). *If $\mathbf{A}$ ($\ell \times m$) and $\mathbf{B}$ ($m \times q$) are constant with respect to $\tilde{\beta}$, and $\tilde{v} = \tilde{v}(\tilde{\beta})$ is a $q \times 1$ vector that depends on the $p \times 1$ vector $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\mathbf{B}\tilde{v})}_{p \times \ell} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{B}^\top}_{q \times m} \underbrace{\mathbf{A}^\top}_{m \times \ell}$$

i Proof

Proof. Apply Theorem 0.5 with the constant $\ell \times q$ matrix $\mathbf{A}\mathbf{B}$, then use the [transpose of a product](#):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\mathbf{B}\tilde{v}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) (\mathbf{A}\mathbf{B})^\top \quad (\text{derivative of a matrix-vector product}) \\ &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{B}^\top \mathbf{A}^\top \quad (\text{transpose of a product}) \end{aligned}$$

□

Example 0.8. Let $\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$ (1×2), $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (2×2), and $\tilde{v}(\tilde{\beta}) = \tilde{\beta}$ where $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{A}\mathbf{B}\tilde{v} = 2\beta_1$, and:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\mathbf{B}\tilde{v})}_{2 \times 1} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right)}_{2 \times 2} \underbrace{\mathbf{B}^\top}_{2 \times 2} \underbrace{\mathbf{A}^\top}_{2 \times 1} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

Corollary 0.1 (Derivative of a dot product, transpose-product form). *If $\tilde{x}$ is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{p \times 1} \left(\underbrace{\tilde{x}^\top}_{1 \times p} \underbrace{\tilde{\beta}}_{p \times 1} \right) = \underbrace{\tilde{x}}_{p \times 1}$$

This vector derivative formula looks a lot like non-vector calculus, except that you have to transpose the coefficient: in scalar calculus $\frac{\partial}{\partial x}(cx) = c$, but here the coefficient $\tilde{x}^\top$ (a row vector) becomes $\tilde{x}$ (a column vector) in the result.

i Proof

Proof. Using Theorem 0.2:

Since $\tilde{x}^\top \tilde{\beta} = \tilde{x} \cdot \tilde{\beta}$ (see [dot product](#)), and $\tilde{x}$ is constant with respect to $\tilde{\beta}$:

$$\frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta}) = \frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta}) = \tilde{x}$$

by Theorem 0.2. □

i Proof

Proof. Using Theorem 0.5:

Since $\tilde{x}$ is constant with respect to $\tilde{\beta}$, $\mathbf{A} = \tilde{x}^\top$ is a constant $1 \times p$ matrix. Applying Theorem 0.5 with $\tilde{v} = \tilde{\beta}$ (so $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}$):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right) (\tilde{x}^\top)^\top \\ &= \mathbf{I} \cdot \tilde{x} \\ &= \tilde{x} \end{aligned}$$

□

Example 0.9 (Derivative of a transpose product). Let $\tilde{x} = (3, 5)^\top$ and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{x}^\top \tilde{\beta} = 3\beta_1 + 5\beta_2$, and by Corollary 0.1:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{2 \times 1} \left(\underbrace{\tilde{x}^\top}_{1 \times 2} \underbrace{\tilde{\beta}}_{2 \times 1} \right) = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Theorem 0.7 (Derivative of a quadratic form). For a quadratic form (see [quadratic form](#)), if $\mathbf{S}$ is a symmetric $p \times p$ matrix that is constant with respect to $\tilde{\beta}$, then:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{p \times 1} = \underbrace{2\mathbf{S}\tilde{\beta}}_{p \times 1}$$

i Proof

Proof. Expanding entry-wise, $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k$. Differentiating component-wise with respect to β_i for $i = 1, \dots, p$:

$$\begin{aligned} \left[\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) \right]_i &= \frac{\partial}{\partial \beta_i} \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k \quad (\text{expand quadratic form}) \\ &= \sum_{k=1}^p s_{ik} \beta_k + \sum_{j=1}^p s_{ji} \beta_j \quad (\text{product rule for } \beta_i \beta_k) \\ &= [\mathbf{S} \tilde{\beta}]_i + [\mathbf{S}^\top \tilde{\beta}]_i \quad (\text{matrix-vector multiplication definition}) \\ &= [(\mathbf{S} + \mathbf{S}^\top) \tilde{\beta}]_i \quad (\text{linearity of matrix multiplication}) \end{aligned}$$

When $\mathbf{S}$ is symmetric ($\mathbf{S} = \mathbf{S}^\top$), $\mathbf{S} + \mathbf{S}^\top = 2\mathbf{S}$, so $\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) = 2\mathbf{S}\tilde{\beta}$. □

This operation is like taking the derivative of cx^2 with respect to x in non-vector calculus.

Example 0.10 (Derivative of a quadratic form). Let $\mathbf{S} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ (2×2 , symmetric and constant) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = 3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2$. By Theorem 0.7:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{2 \times 1} = 2\mathbf{S}\tilde{\beta} = 2 \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Differentiating component-wise directly:

$$\begin{pmatrix} \frac{\partial}{\partial \beta_1}(3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \\ \frac{\partial}{\partial \beta_2}(3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Both methods agree.

Corollary 0.2 (Derivative of a simple quadratic form).

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{p \times 1} (\tilde{\beta}^\top \tilde{\beta}) = \underbrace{2\tilde{\beta}}_{p \times 1}$$

i Proof

Proof. Applying Theorem 0.7 with $\mathbf{S} = \mathbf{I}_{p \times p}$ (which is symmetric and constant with respect to $\tilde{\beta}$):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \tilde{\beta}) &= \frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{I}_{p \times p} \tilde{\beta}) \quad (\text{rewrite with identity matrix}) \\ &= 2\mathbf{I}_{p \times p} \tilde{\beta} \quad (\text{derivative of a quadratic form, with } \mathbf{S} = \mathbf{I}_{p \times p}) \\ &= 2\tilde{\beta} \quad (\text{identity matrix property}) \end{aligned}$$

□

This vector derivative is like taking the derivative of x^2 .

Example 0.11 (Derivative of a sum of squares). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$, so $\tilde{\beta}^\top \tilde{\beta} = \beta_1^2 + \beta_2^2$. By Corollary 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{2 \times 1} (\tilde{\beta}^\top \tilde{\beta}) = 2\tilde{\beta} = \begin{pmatrix} 2\beta_1 \\ 2\beta_2 \end{pmatrix}$$

Direct partial differentiation yields the same column vector.

Theorem 0.8 (Vector chain rule). *Let $\tilde{x}$ be a $p \times 1$ vector, let $\tilde{y} = \tilde{g}(\tilde{x})$ be a $q \times 1$ vector-valued function of $\tilde{x}$, and let $z = f(\tilde{y})$ be a scalar-valued function of $\tilde{y}$, where $\tilde{g}$ and f have continuous partial derivatives. Then $z = f(\tilde{g}(\tilde{x}))$, as a function of $\tilde{x}$, satisfies*

$$\underbrace{\frac{\partial z}{\partial \tilde{x}}}_{p \times 1} = \underbrace{\frac{\partial \tilde{y}}{\partial \tilde{x}}}_{p \times q} \underbrace{\frac{\partial z}{\partial \tilde{y}}}_{q \times 1}$$

where $\frac{\partial \tilde{y}}{\partial \tilde{x}}$ is the derivative of Definition 0.3 and $\frac{\partial z}{\partial \tilde{x}}$ and $\frac{\partial z}{\partial \tilde{y}}$ are vector derivatives (Definition 0.1).

See <https://quickfem.com/finite-element-analysis/>, specifically https://quickfem.com/wp-content/uploads/IFEM.AppF_.pdf

See also https://en.wikipedia.org/wiki/Gradient#Relationship_with_Fr%C3%A9chet_derivative

This chain rule is like the univariate [chain rule](#), but the order matters now. The version presented here is for the [gradient](#) (column vector); the [total derivative](#) (row vector) would be the [transpose of the gradient](#).

Corollary 0.3 (Vector chain rule for quadratic forms). *If $\tilde{\varepsilon} = \tilde{\varepsilon}(\tilde{\beta})$ is an $n \times 1$ vector-valued function of the $p \times 1$ vector $\tilde{\beta}$, with continuous partial derivatives, then*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon}(\tilde{\beta}) \cdot \tilde{\varepsilon}(\tilde{\beta}))}_{p \times 1} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon}(\tilde{\beta}) \right)}_{p \times n} \underbrace{(2\tilde{\varepsilon}(\tilde{\beta}))}_{n \times 1}$$

i Proof

Proof. Apply Theorem 0.8 with $\tilde{x} = \tilde{\beta}$, $\tilde{y} = \tilde{\varepsilon}$, $q = n$, and $z = f(\tilde{\varepsilon}) = \tilde{\varepsilon} \cdot \tilde{\varepsilon} = \tilde{\varepsilon}^\top \tilde{\varepsilon}$:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) \frac{\partial}{\partial \tilde{\varepsilon}}(\tilde{\varepsilon}^\top \tilde{\varepsilon}) \quad (\text{vector chain rule}) \\ &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) (2\tilde{\varepsilon}) \quad (\text{derivative of a simple quadratic form, in } \tilde{\varepsilon}) \end{aligned}$$

The second step is Corollary 0.2, applied to the $n \times 1$ vector $\tilde{\varepsilon}$ in place of $\tilde{\beta}$. □

Example 0.12 (Derivative of the residual sum of squares). Let $\tilde{y}$ ($n \times 1$) and $\mathbf{X}$ ($n \times p$) be constant with respect to $\tilde{\beta}$, and let $\tilde{\varepsilon}(\tilde{\beta}) = \tilde{y} - \mathbf{X}\tilde{\beta}$ be the vector of residuals. By Theorem 0.4, $\frac{\partial}{\partial \tilde{\beta}}(\mathbf{X}\tilde{\beta}) = \mathbf{X}^\top$, and $\frac{\partial}{\partial \tilde{\beta}}\tilde{y} = \mathbf{0}_{p \times n}$ because $\tilde{y}$ is constant, so $\frac{\partial}{\partial \tilde{\beta}}\tilde{\varepsilon} = -\mathbf{X}^\top$ ($p \times n$). By Corollary 0.3:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= (-\mathbf{X}^\top)(2\tilde{\varepsilon}) \quad (\text{vector chain rule for quadratic forms}) \\ &= -2\mathbf{X}^\top(\tilde{y} - \mathbf{X}\tilde{\beta}) \quad (\text{substitute } \tilde{\varepsilon} = \tilde{y} - \mathbf{X}\tilde{\beta}) \end{aligned}$$

Setting this $p \times 1$ vector to $\tilde{0}$ gives the normal equations $\mathbf{X}^\top \mathbf{X} \tilde{\beta} = \mathbf{X}^\top \tilde{y}$ of least squares.

Definition 0.5 (Trace). The **trace** of a $p \times p$ matrix $\mathbf{M}$ is the sum of its diagonal entries:

$$\text{tr}(\mathbf{M}) \stackrel{\text{def}}{=} \sum_{i=1}^p M_{ii}$$

Example 0.13 (The trace of a 2×2 matrix).

$$\text{tr}\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right) = 1 + 4 = 5$$

Definition 0.6 (Matrix derivative). For a scalar-valued function $f(\mathbf{X})$ of an $m \times n$ matrix $\mathbf{X}$, the **matrix derivative** is the $m \times n$ matrix whose (i, j) entry is the partial derivative of f with respect to the (i, j) entry of $\mathbf{X}$:

$$\left[\frac{\partial}{\partial \mathbf{X}} f\right]_{ij} = \frac{\partial}{\partial X_{ij}} f$$

Example 0.14 (The matrix derivative of a trace). Let $\mathbf{X}$ be a 2×2 matrix and $f(\mathbf{X}) = \text{tr}(\mathbf{X}) = X_{11} + X_{22}$ (Definition 0.5). Then $\frac{\partial}{\partial X_{ij}} f = 1$ if $i = j$ and 0 otherwise, so:

$$\frac{\partial}{\partial \mathbf{X}} f = \mathbf{I}_2$$

Theorem 0.9 (Matrix derivative of the trace of a matrix product). If $\mathbf{A}$ ($r \times m$) and $\mathbf{B}$ ($n \times r$) are constant with respect to the $m \times n$ matrix $\mathbf{X}$, then:

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB})}_{m \times n} = \underbrace{\mathbf{A}^\top}_{m \times r} \underbrace{\mathbf{B}^\top}_{r \times n}$$

The trace makes $\text{tr}(\mathbf{AXB})$ a scalar, so its matrix derivative is again an $m \times n$ matrix. The derivative of the matrix product $\mathbf{AXB}$ itself (without the trace) is a fourth-order tensor, which is why this result is stated for the scalar $\text{tr}(\mathbf{AXB})$.

i Proof

Proof. Write A_{kl} , X_{kl} , and B_{kl} for the entries of $\mathbf{A}$, $\mathbf{X}$, and $\mathbf{B}$. For entry (i, j) :

$$\begin{aligned}
\left[\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB}) \right]_{ij} &= \frac{\partial}{\partial X_{ij}} \sum_{a=1}^r \sum_{b=1}^m \sum_{c=1}^n A_{ab} X_{bc} B_{ca} \quad (\text{trace of the } r \times r \text{ product } \mathbf{AXB}) \\
&= \sum_{a=1}^r A_{ai} B_{ja} \quad (\text{only the terms with } b = i, c = j \text{ depend on } X_{ij}) \\
&= \sum_{a=1}^r [\mathbf{A}^\top]_{ia} [\mathbf{B}^\top]_{aj} \quad (\text{definition of the transpose}) \\
&= [\mathbf{A}^\top \mathbf{B}^\top]_{ij} \quad (\text{definition of matrix multiplication})
\end{aligned}$$

□

Example 0.15 (Differentiating a weighted trace). Let $\mathbf{A} = \mathbf{I}_2$ (2×2) and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (2×2). Then $\text{tr}(\mathbf{AXB}) = 2X_{11} + 3X_{22}$, and:

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB})}_{2 \times 2} = \underbrace{\mathbf{A}^\top}_{2 \times 2} \underbrace{\mathbf{B}^\top}_{2 \times 2} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

Additional resources

See also the [Linear Algebra and Vector Calculus references](#).

- [Hua Zhou's lecture notes for "UCLA Biostat 216 - Mathematical Methods for Biostatistics" \(2023 Fall\)](#)

References

Fieller, Nick. 2016. *Basics of Matrix Algebra for Statistics with R*. Chapman; Hall/CRC. <https://doi.org/10.1201/9781315370200>.